SELF-EVOLUTION SHOULD NOT DEPEND ON HOW SKILLS ARE SPLIT: AN EXACT CERTIFICATE FOR SKILL-TAXONOMY REPRESENTATION INVARIANCE

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ABSTRACT

Self-evolving language-model agents increasingly use skill identities as control objects: a sampled skill can determine the next training task, a dynamic ID can receive retrieval priority, and feedback is often written back to the selected package. This creates a basic representation question: if two skill libraries expose the same semantic capabilities but differ only in how those capabilities are split, cloned, or grouped into package identities, should the induced self-evolution control change? We formulate *Skill-Taxonomy Representation Invariance* (STRI) and show that released self-evolving skill systems can violate it. In Skill Self-Play (Skill-SP), 183 of 314 released-validator-covered API-Bank Level-1 tasks have multiple skill memberships; the release's text duplicate filter misses all observed specific–generic support-overlap pairs, and exact-support package pruning still leaves 71 overlapping tasks. In SkillRL, exact-content skills with fresh dynamic IDs are accepted 12/12 times, and 11/12 such identity changes alter the unique semantic contents returned under a fixed retrieval budget. For a frozen context-by-package support matrix A, define $R^*(A) = \min_{w \geq 0, t \geq 0}\{t : \mathbf{1} \leq Aw \leq t\mathbf{1}\}$. We prove that $R^*(A) = 1$ iff nonnegative package weights can equalize additive semantic exposure over all covered contexts, while $R^*(A) > 1$ is the tight minimum multiplicative distortion. On Skill-SP API-Bank Level-1, $R^* = 2$; the same value independently holds on a frozen calibration subset and a tool-disjoint heldout subset. Two real negative regimes sharpen the boundary: Level-3 has $R^* = 1$ with disjoint support, while Skill-SP's released logical compilers yield 127/128 multi-membership tasks yet still have $R^* = 1$ because an umbrella support permits exact equalization. Thus overlap prevalence is not the cause; *non-equalizable support geometry* is. Explicit clone/split ablations further show why quotienting matters: raw package multiplicity and even the closed-form singleton witness can change under an exact duplicate representation, while exact-support quotienting recovers every audited support row and the original witness count. The positive Level-1 witness also survives every leave-one-tool and leave-one-row deletion. We operationalize this result as STRI-Cert, a fail-closed audit protocol that computes the exact certificate and optional human-readable lower-bound witnesses from independently defined support contracts. We do not claim LP computational novelty or downstream task harm. A faithful Qwen3-8B dynamic pilot completed its frozen generation budget but failed the author-native qualification gate, so it provides no dynamic scientific update and is not rerun.

1 INTRODUCTION

Many self-evolving agents adapt through external state—memories, skills, workflows, tools, prompts, retrieval policies, or harness code—rather than only through foundation-model weights. Skills are especially attractive because a skill can package reusable procedure, metadata, examples, applicability information, and verification logic. Recent systems make the skill identity itself a control variable: Skill-SP (Huang et al., 2026) samples a skill before generating a training task, while SkillRL (Xia et al., 2026) gives dynamically created skill IDs priority in a fixed retrieval budget.

This architecture makes the *representation* of the skill library consequential. Consider two taxonomies that expose the same executable semantic capability. One stores a primitive once; another stores an exact clone under a fresh ID. One represents generic and specific capabilities separately; another groups unchanged primitives behind a macro package. If no new primitive behavior or support region is added, should the future curriculum, retrieval exposure, or credit allocation change solely because package bookkeeping changed?

Existing work on semantic skill clones, redundancy governance, skill safety, adaptive routing, compression, and structured bandits establishes that skill quality, redundancy, and routing matter. These lines do not define a representation-invariance property for an endogenous evolution controller. We therefore ask: *when can package-indexed control be made invariant to semantics-preserving changes in the skill taxonomy using the same frozen support information?*

We approach the question asset-first. Rather than proposing a normalization method and manufacturing a benchmark, we inspect released control surfaces, reduce the phenomenon against strongest same-information baselines, and identify a support geometry that separates positive and negative regimes.

Released Skill-SP. On API-Bank Level-1, 314 released tasks are covered by at least one released validator and 183 have multiple memberships. The release's name+description Jaccard duplicate filter at threshold 0.33 misses all five observed specific–generic support-overlap pairs. Minimum-cardinality whole-package pruning can reduce the active library from 10 to 7 while preserving exact support, but 71 overlapping tasks remain. The residual is therefore not merely an exact-clone artifact.

Released SkillRL. In released template-mode retrieval, dynamic general skills are included before the remaining fixed `top_k=6` budget is filled with static skills. Adding one exact-content fresh-ID copy at a time admits all 12 tested clones. Across 223 released ALFWorld task descriptions, 11/12 clone targets change the unique semantic retrieval set and 5/12 reduce the number of unique general-skill contents returned.

These systems establish representation sensitivity, but not its mechanism. A natural hypothesis is simply that more overlap causes more sensitivity. Our main result rejects that hypothesis.

Contributions.

1. **STRI problem and evidence.** We formulate Skill-Taxonomy Representation Invariance and document package-identity representation sensitivity in two independent released self-evolving skill systems.
2. **Exact certificate.** We derive $R^*(A)$, an exact iff criterion for whether nonnegative package weights can equalize additive semantic exposure on a frozen support matrix, plus a global-singleton overlap theorem giving an interpretable factor-2 lower bound.
3. **Real mechanism boundary.** Skill-SP Level-1, a frozen calibration subset, and a tool-disjoint heldout subset all have $R^* = 2$; Level-3 has $R^* = 1$; and 127/128 first-party logical-compiler tasks overlap while still yielding $R^* = 1$. This falsifies overlap prevalence as the mechanism.
4. **Matched baselines and ablations.** We compare raw overlap, exact support-signature deduplication, minimum exact-coverage pruning, and the exact best global reweighting on identical support information; we then stress the diagnosis with exact clone/split, identity-renaming, leave-one-tool, leave-one-row, and tool-resampling controls.
5. **STRI-Cert.** We package the criterion and witness search into a fail-closed audit protocol without claiming a new LP algorithm, downstream improvement, or a validated dynamic repair.

2 From Package Identity to Semantic Exposure

2.1 Frozen support matrix

Let $X = \{x_1, \ldots, x_n\}$ be a finite set of semantic contexts or tasks with independently defined first-party support truth, and let $S = \{s_1, \ldots, s_m\}$ be the current skill packages. Define the binary

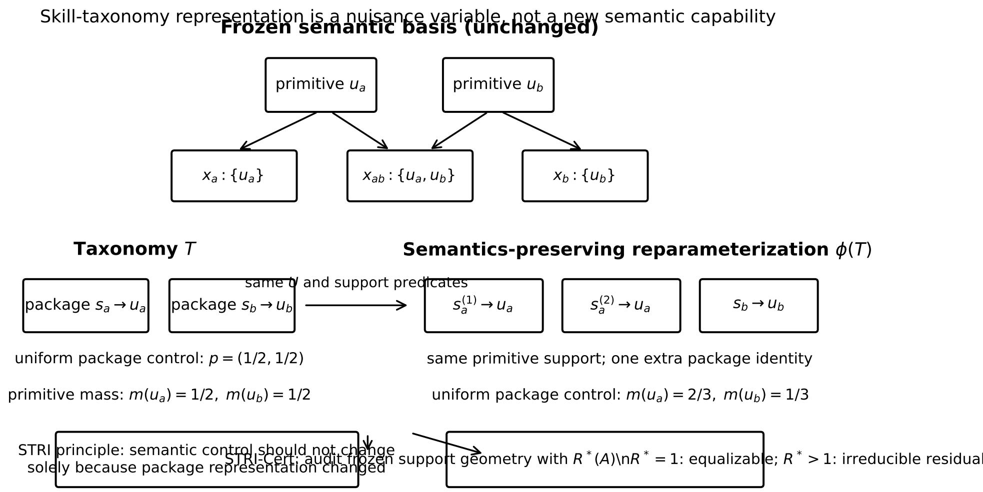

Figure 1: STRI separates semantic capability from package representation. The conceptual identity-split example leaves the primitive basis and support predicates unchanged, yet a package-indexed controller can change primitive mass solely because one primitive has two package identities. Exact clones motivate the nuisance channel; the decisive released residual in Sec. 4 survives clone reasoning and is characterized by non-equalizable partial-overlap support geometry.

incidence matrix

$$A_{ij} = \mathbf{1}\{s_j \text{ supports } x_i\}.$$

The support definition is frozen before evaluating STRI. In the tool-call study, support comes from released Skill-SP validators. In the logical-reasoning study, tasks are generated by released compiler specifications, checked by the author's fixed CSP/contract code, and cross-skill support is determined by the released compiler-alignment predicate.

We study package-only pre-context control. At a frozen state, the controller chooses nonnegative package mass $w \in \mathbb{R}^m_{\geq 0}$ before the semantic task to be generated exists. The additive eligibility exposure of context x_i is $e_i = (Aw)_i$. This is a control-opportunity quantity, not a task probability and not utility.

2.2 Same-information pre-task controllers reduce to package mass

In released Skill-SP, skill selection occurs before task generation. At this decision point the action is one current package identity. For any fixed pre-task state z, every randomized same-information rule—including a bandit, MLP, or neural router—induces a categorical distribution over packages, hence one nonnegative mass vector $w(z)$. The strongest same-information baseline is therefore arbitrary nonnegative package mass with access to the complete frozen support matrix.

This reduction is pointwise in z. It does *not* claim that a longitudinal adaptive controller is globally equivalent to one static vector, and it does not allow a baseline to condition on the future generated task when the reviewed controller acts before that task exists.

3 Exact Equalizability and STRI-Cert

3.1 Exact finite-support criterion

The max/min exposure ratio is invariant to positive scaling of w. We normalize every covered context to exposure at least one and define

$$R^*(A) = \min_{w \geq 0,\, t \geq 0} t \quad \text{s.t.} \quad \mathbf{1} \leq Aw \leq t\mathbf{1}. \tag{1}$$

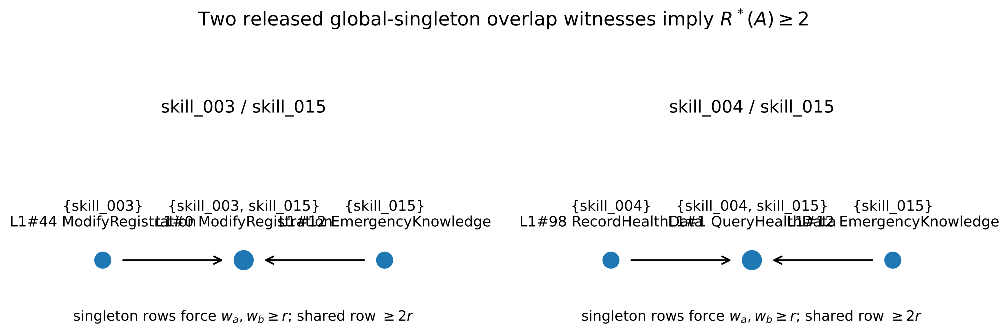

Figure 2: Two released global-singleton overlap witnesses. In each case, retaining positive exposure on the two singleton regions forces at least twice that minimum exposure on the shared region.

Theorem 1 (finite-support package-only equalizability). $R^*(A) = 1$ if and only if there exists a nonnegative package-only controller that gives exactly equal additive exposure to every covered context. If $R^*(A) > 1$, no such package weighting exists, and $R^*(A)$ is the tight minimum achievable max/min exposure ratio.

Proof. If exact equalization exists, rescale its positive common exposure to one, giving a feasible solution with $t = 1$. The constraints imply $t \geq 1$, so $R^*(A) = 1$. Conversely, $R^*(A) = 1$ implies $\mathbf{1} \leq Aw \leq \mathbf{1}$, hence $Aw = \mathbf{1}$. For any w with positive exposure on every covered row, let $r = \min_i(Aw)_i$ and rescale to $\tilde{w} = w/r$. Then $\min_i(A\tilde{w})_i = 1$ and the smallest feasible t is $\max_i(Aw)_i / \min_i(Aw)_i$. Minimizing over w yields exactly the tight multiplicative distortion. $\square$

The optimization itself is standard linear programming. The scientific content is the representation-invariance object that the LP exactly decides for a released self-evolution control surface.

3.2 Human-readable factor-2 witness

Theorem 2 (global-singleton overlap lower bound). Suppose packages a, b each have a context supported by exactly that one package over the full frozen support universe, and another context is supported by both a and b, possibly with additional packages. If both singleton contexts retain positive exposure, then every nonnegative package mass has max/min exposure ratio at least two on the focal rows, and therefore $R^*(A) \geq 2$ for the full support matrix.

Proof. Let $r > 0$ be the minimum exposure among the two singleton contexts and the shared context. Since the first singleton support signature is exactly $\{a\}$, its exposure is $w_a \geq r$; likewise $w_b \geq r$. The shared context contains both a and b, and additional packages have nonnegative weight, so its exposure is at least $w_a + w_b \geq 2r$. Thus every w has focal max/min ratio at least two. For the same w, the full-matrix maximum is no smaller than the focal maximum and the full-matrix minimum is no larger than the focal minimum, so the full-matrix ratio is also at least two. Minimizing over w gives $R^*(A) \geq 2$. $\square$

The released Skill-SP Level-1 graph contains two independent instances, involving `skill_003`–`skill_015` and `skill_004`–`skill_015`. The exact LP attains $R^* = 2$, so the lower bound is tight.

3.3 STRI-Cert

STRI-Cert is a diagnostic protocol rather than a learned predictor or a new LP solver:

1. freeze an independently defined context-by-package support matrix;
2. solve Eq. (1);
3. if $R^* = 1$, report *package-only equalizable*;

Table 1: Released-system evidence for package-identity representation sensitivity. The two systems expose different control channels; only Skill-SP supplies the support matrix used by STRI-Cert.

System	Control surface	Released evidence
Skill Self-Play (tool-call)	Pre-task skill sampling and skill-ID feedback/refinement	183 multi-membership Level-1 rows among 314 covered rows; the released text duplicate filter misses all 5/5 observed specific–generic support-overlap pairs; exact-support whole-package pruning still leaves 71 overlap rows.
SkillRL	Fixed-budget template-mode dynamic skill retrieval	12/12 exact-content fresh-ID clones are admitted; 11/12 clone targets change the unique semantic retrieval set over 223 released ALFWorld task descriptions; 5/12 reduce the number of unique general-skill contents.

Table 2: Matched baseline ladder on the Skill-SP tool-call support matrix. Every baseline uses the same frozen released support information; the ladder deliberately strengthens from name-level duplicate removal to the exact best context-independent package reweighting.

Baseline / reduction	Outcome	Evidence
Released name+description Jaccard duplicate filter	Does not absorb	Misses all 5/5 observed specific–generic support-overlap pairs; similarity 0.1429–0.20 is below the released 0.33 threshold.
Exact semantic-clone deduplication	Does not absorb partial overlap	The positive regime uses distinct specific/generic packages with different unique support regions.
Minimum-cardinality whole-package pruning preserving exact support	Partial reduction only	Active packages drop from 10 to 7, but 71 overlap rows remain while exact support is preserved.
Arbitrary nonnegative global package weights	Exact residual	The exact optimum is $R^*(A) = 2$.
Global-singleton overlap witness	Closed-form lower bound	Two independent released witnesses imply $R^*(A) \geq 2$, and the exact LP attains 2.

4. if $R^* > 1$, report an *irreducible package-only representation residual* and its tight distortion;
5. search for globally valid singleton-overlap structures to provide interpretable lower bounds when available;
6. fail closed if support truth or optimization is invalid.

The strongest same-information reduction is built into the certificate: arbitrary nonnegative global package reweighting receives the complete frozen support matrix. Text deduplication, exact-clone removal, uniform routing, and fixed scalar retuning are weaker reductions.

4 Released-System Evidence

4.1 Skill-SP tool-call support: positive regime

On API-Bank Level-1, 314 rows are accepted by at least one released Skill-SP validator and 183 have multiple memberships. STRI-Cert gives $R^* = 2.0$.

To test whether this is an artifact of fitting package weights to the complete table, we use a tool-disjoint split frozen before the later certificate analysis. Calibration uses `AppointmentRegistration`, `ModifyRegistration`, `QueryHealthData`, and `RecordHealthData`; heldout uses `CancelRegistration`, `EmergencyKnowledge`, `QueryRegistration`, and `SymptomSearch`. Calibration has 47 covered tasks, 33 multi-membership rows, and $R^* = 2$. Heldout has 52 covered tasks, 38 multi-membership rows, and again $R^* = 2$. Both independently contain the two globally valid factor-2 witness structures.

Table 3: Mechanism boundary across real first-party support regimes. High overlap is neither necessary nor sufficient for an irreducible package-only residual; the exact discriminator is $R^*(A)$.

Regime	Covered	Multi	Overlap	R^*	Certificate
API-Bank Level-1 full	314	183	58.3%	2.0	Residual
Level-1 calibration tools	47	33	70.2%	2.0	Residual
Level-1 tool-disjoint heldout	52	38	73.1%	2.0	Residual
API-Bank Level-3	34	0	0.0%	1.0	Equalizable
Logical compiler validation	128	127	99.2%	1.0	Equalizable

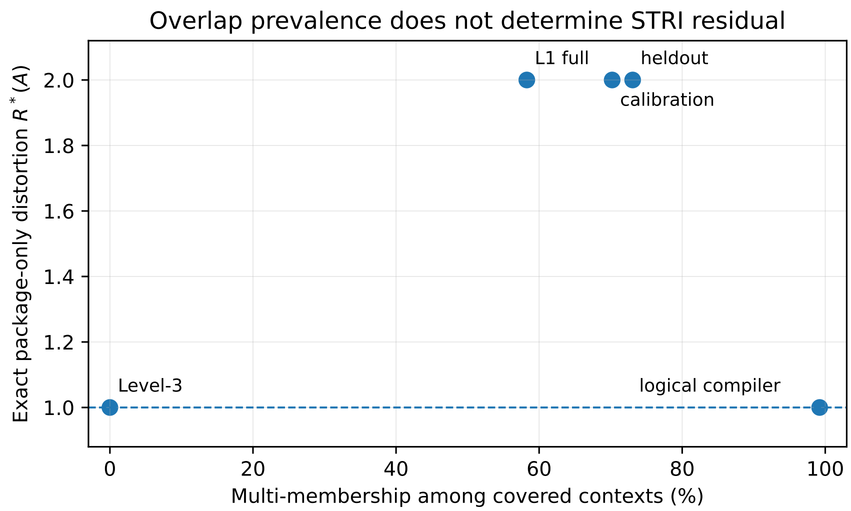

Figure 3: Overlap prevalence does not determine STRI residual. The three Level-1 regimes independently have $R^*(A) = 2$, while Level-3 is equalizable with disjoint support and the logical compiler domain is equalizable despite 127/128 multi-membership tasks.

The witness rows are directly inspectable. For `skill_003`/`skill_015`, L1#44 and L1#12 are singleton regions while L1#0 is shared. For `skill_004`/`skill_015`, L1#98 and L1#12 are singleton regions while L1#1 is shared. In each case Theorem 2 yields the same lower bound attained by the exact LP.

4.2 Level-3: Disjoint-support negative control

Level-3 provides a natural released negative regime. Thirty-four rows are covered, every covered row has exactly one skill membership, and $R^* = 1$. Equal weights on the active packages equalize every covered task. STRI-Cert therefore does not mechanically label every released taxonomy problematic.

4.3 Logical compiler domain: high overlap but no residual

A stronger negative test asks whether high overlap alone triggers the certificate. Skill-SP's logical-reasoning release contains eight skill packages with programmatic compiler specifications. The author code can generate a zebra puzzle from a compiler, verify it with a fixed CSP solver, and check whether a generated sample satisfies each package's released compiler-alignment contract.

We freeze the author's default compiler-validation sizes $(3\times3, 4\times4, 5\times5, 6\times6)$ and seeds $0, 1, 2, 3$ for each of eight released skills, giving 128 source units. No language model is used. All 128/128 units compile and pass the author's puzzle contract and uniqueness verifier; every task is then cross-evaluated against all eight released support specifications.

The support matrix is extremely overlapping: 127/128 tasks have more than one supported skill, with 38 distinct support patterns. Yet $R^* = 1$. The exact LP places weight one on `skill_008` and zero on the remaining packages because the released `skill_008` alignment contract supports every generated unit. This umbrella support makes the high-overlap matrix exactly equalizable.

Thus the mechanism is not “more overlap implies more representation dependence.” It is whether the support geometry admits an equalizing nonnegative package mixture.

4.4 SkillRL: independent identity-sensitivity witness

SkillRL shows that the broader representation-sensitivity problem is not confined to one implementation. In released template-mode retrieval, fresh dynamic identities receive priority in a fixed general-skill budget. An exact-content clone under a fresh `dyn_NNN` ID is admitted for all 12 tested general skills. For 11/12 clone targets, the unique semantic retrieval set changes over 223 released task descriptions; for 5/12, the number of unique general contents decreases under the same `top_k=6` budget.

We use SkillRL only as an independent identity-sensitivity witness. The exact R^* analysis is performed on Skill-SP, where first-party support contracts expose the required support matrix directly.

4.5 Matched baselines, representation ablations, and failure analysis

Table 2 is an explicit same-information baseline ladder, not a collection of unrelated heuristics. The first three baselines may inspect the same complete released support object but can only remove duplicate package columns or whole packages. Minimum exact-coverage pruning is already strong: it removes 61.2% of the observed Level-1 overlap (183 to 71 multi-membership rows). Yet the residual survives. The strongest context-independent baseline then receives arbitrary nonnegative package weights and the full support matrix; its exact optimum remains $R^* = 2$ on the full, calibration, and heldout Level-1 regimes. This is the point at which a simpler package-level explanation is exhausted.

We next ablate the representation itself (Figure 4A). Each active Level-1 package is cloned and split independently while preserving its support column. These interventions deliberately perturb package multiplicity without adding a support region. Before quotienting, cloning `skill_003`, `skill_004`, or the generic `skill_015` changes the two-witness closed-form diagnostic to 1, 1, and 0 witnesses respectively. This is not a change in the exact certificate: duplicating a support column leaves R^* unchanged because duplicate weights can be merged or one duplicate can receive zero mass. It instead exposes an implementation pitfall in the human-readable witness search. Exact-support quotienting removes the nuisance duplication and recovers every original support row and both witnesses for every clone and split control. A bijective renaming of all package IDs leaves the diagnostics unchanged. Thus the quotient is not cosmetic; it is required for the interpretable witness layer to inherit the representation invariance of the exact LP.

Finally, we ask where the simple witness succeeds and fails. Across the 49 Level-1 tools, 4 have a closed-form witness locally, 10 contain overlap but no singleton-overlap witness, 20 are disjoint/singleton-only, and 15 have no covered support (Figure 4C). We mark the 10 overlap-without-witness tools *inconclusive*, not equalizable: absence of Theorem 2’s sufficient structure is not a negative certificate, so those cases must defer to the exact LP. At the aggregate Level-1 scale, both released factor-2 witnesses survive all 49 leave-one-tool deletions and all 399 leave-one-row deletions. In 500 fixed-seed tool-bootstrap resamples, at least one witness remains in 100% of replicates and both remain in 97.2% (Figure 4B). The corresponding multi-membership fraction spans 35.1–73.3% from the 5th to 95th percentile, yet the witness is stable. We treat this bootstrap only as descriptive subset robustness: the frozen support counts themselves are exact finite-snapshot quantities, not estimates of a population-wide prevalence.

5 Relation to Existing Work

Skill redundancy and clone detection. Prior work studies semantic skill clones, duplicate propagation, and clone-aware filtering (Zhu et al., 2026). STRI therefore does not claim that duplicate skills exist or that exact duplicates should be detected. The focal positive regime survives exact-clone reasoning because specific and generic packages have distinct unique support regions in addition to shared contexts.

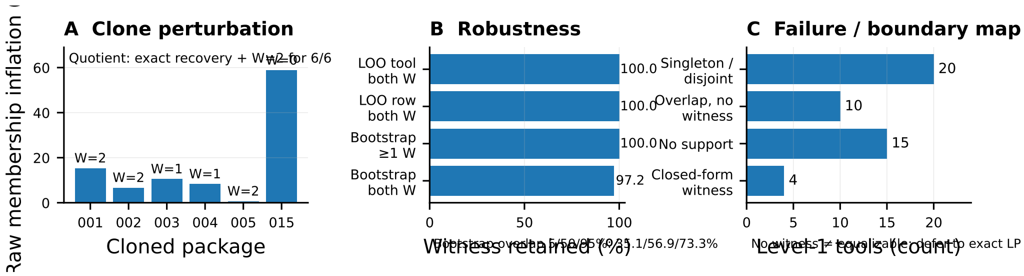

Figure 4: Representation-ablation and robustness evidence. **A:** clone-induced raw multiplicity/witness changes vanish under exact-support quotienting. **B:** the factor-2 witnesses survive deletion tests and 500 tool resamples. **C:** witness absence is fail-closed: overlap without a witness is inconclusive and defers to the exact LP.

Redundancy-aware lifecycle governance. Prior governance methods recommend, attribute, suppress, merge, or update redundant and environment-sensitive skills (Liu et al., 2026). Such policies may improve performance, but they do not answer whether a frozen taxonomy admits representation-invariant package-only control. STRI-Cert is a property test of control/support geometry rather than a competing recommendation algorithm.

Skill safety and failures. Work on unsafe skill evolution and harmful agent skills demonstrates that learned or retrieved skill content can propagate functional or efficiency failures (Mao et al., 2026; Dong et al., 2026). Our primary support transformations keep the audited semantic support predicates fixed. A positive STRI certificate does not imply unsafe content or downstream failure.

Adaptive routing and evolving skills. ERSkill, Skill-SP, SkillRL, SHAPER, and related systems learn routers or co-evolve skill libraries (Chen et al., 2026; Huang et al., 2026; Xia et al., 2026; Wang et al., 2026). These works absorb any generic claim that adaptive routing or skill evolution is novel. They motivate STRI because package identities become endogenous actions or state, but they do not test invariance under a semantics-preserving reparameterization of that representation.

Compression and structured online learning. Skill-compression methods factor repeated procedural structure (Bai et al., 2026), while classical structured bandit/expert methods share information across similar fixed actions (Slivkins, 2011; Sen et al., 2018). We concede both mature ideas. STRI is not a new optimizer over similar arms; it is the exact representation-invariance question for an endogenous artifact taxonomy and a certificate on independently defined semantic support.

6 A PREREGISTERED DYNAMIC PILOT THAT DID NOT QUALIFY

Static support distortion need not propagate through a real task generator. We therefore preregistered a faithful Qwen3-8B Skill-SP pilot before using any dynamic outcome. The contract fixed three source skills (003, 004, 015), 24 generations per source, author prompts and validators, released-style decoding, and a minimum of 16 contract-valid tasks per source before any merge/split statistic could update belief.

All 72 frozen generations completed within budget. However, only 14/24, 5/24, and 8/24 outputs for the three sources satisfied the author-native task contract. The formal outcome is `INCONCLUSIVE_PROPOSER_QUALIFICATION_FAILED`; no dynamic witness statistic is scientifically admissible.

The dominant failures are requirements implemented by the author questioner contract itself, especially omission of the required solver <tool_call>...</tool_call> output contract. The released training pipeline uses the same contract checks before solver evaluation. We therefore classify the outcome as proposer/substrate competence insufficiency rather than a negative STRI result or a wrong observable. We do not lower the 16/source gate, add generations, change decoding, rerun the same contract, or switch backbone after observing the qualification failure.

We also preregistered a SkillRL fixed-task bridge with the author's final ALFWorld policy. A competence stage reached 18/24 success across five task families. The subsequent 24 paired A/B/C/D units gave 18/24 terminal success in every arm and zero paired endpoint disagreements: exact-clone semantic displacement B changed action trajectories in 11/24 units, while the matched identity placebo C changed 15/24, and exact quotienting D restored A trajectories in 24/24. Thus this qualified realization stops the proposed exact-clone-to-final-success bridge; it does not establish a population-level no-effect theorem because no equivalence margin or task-cluster power analysis was preregistered. This optional C4 result cannot rescue or invalidate the N1–N3 support-geometry claims.

7 Discussion and Claim Boundary

STRI-Cert answers a precise question: given an independently defined finite support matrix and package-only nonnegative mass, is exact additive exposure equality achievable? A positive certificate ($R^* > 1$) says that every package-only global weighting leaves a representation-dependent exposure distortion on the audited support domain. It does not say which package should be deleted, that the taxonomy is semantically bad, or that downstream utility decreases. A negative certificate ($R^* = 1$) says that the support geometry is equalizable by some package mixture; it does not guarantee that the released controller actually uses that mixture or that all longitudinal dynamics are invariant.

The logical-compiler result illustrates why this distinction matters. Its 99.2% overlap rate appears more severe than the tool-call regime if one counts redundancy. STRI-Cert reaches the opposite mechanistic conclusion because an umbrella package makes exact equalization possible. A redundancy count or text-similarity score misses this boundary.

The narrow claims are intentionally smaller than our original dynamic design. We **do** claim that released control surfaces can depend on skill-package representation, that $R^*(A)$ exactly characterizes package-only additive exposure equalizability on a frozen finite support matrix, and that the audited residual tracks support geometry rather than overlap prevalence. We **do not** claim that a positive certificate causes task failure, that static exposure predicts longitudinal utility, that STRI-Cert is computationally novel relative to linear programming, that high overlap is harmful in general, that Support-Quotient Control has been empirically validated, that the unqualified Qwen3 bank provides dynamic evidence, or that the qualified SkillRL bridge STOP proves population-level absence of a downstream effect.

A natural repair is to factor control through semantic support cells; testing such a utility-improving redesign is future work. Overall, STRI-Cert isolates support geometry—not overlap prevalence—as the audited boundary without implying downstream utility.

8 Limitations

1. **Support truth is required.** STRI-Cert assumes a frozen, independently defined support matrix. Learned or uncertain support estimators introduce a separate identification problem.
2. **Static exposure is not utility.** R^* does not establish realized task-distribution change, regret, success degradation, or end-of-evolution harm.
3. **Finite audited contexts.** The certificate is exact on the observed support matrix, not on unobserved semantic contexts outside the audited domain.
4. **Package-only action class.** The criterion addresses controllers whose reviewed pre-context action is package mass. A system that acts directly on semantic support cells may avoid this representation channel by construction.
5. **Dynamic evidence remains bounded.** The faithful Skill-SP Qwen3 pilot failed its preregistered proposer gate. The qualified SkillRL fixed-task bridge produced no terminal-success disagreement beyond placebo, but was not designed as an equivalence/power study. Neither result expands the N1–N3 claim scope.

AI USE STATEMENT

In this work, generative AI tools assisted with conceptual framing and hypothesis refinement; formulation and critique of mathematical claims and proof drafts; experiment and falsifier design; implementation and review of analysis, experiment-control, and manuscript-quality-assurance code; interpretation of intermediate results; primary-source literature triage; and preparation or editing of manuscript text, figures, tables, and references. Generative models were also used as the task proposer in the preregistered Qwen3-8B dynamic pilot reported in Section 6; that pilot failed its frozen author-native qualification gate, and its generated bank was excluded from scientific claims about dynamic STRI. Load-bearing numerical claims in the submitted narrow paper are bound to released first-party artifacts or deterministic/programmatic analysis outputs. The support certificate, theorem edge cases, transition logic, and paper-level numerical/citation constraints were exercised by executable tests; related-work boundaries and bibliographic metadata were checked against primary or official sources. AI-assisted outputs were not treated as independent scientific authority. We take responsibility for the final content of this work, including text, claims, code, and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The main paper states the finite-support assumptions and complete proofs needed for the two certificate theorems, identifies the released-system support regimes used for every load-bearing result, and reports the exact positive and negative mechanism boundaries. The supplementary material will include the STRI-Cert implementation, first-party logical-compiler support protocol, frozen result/contract receipts, source ledger, figure-generation scripts, and machine-readable claim-to-evidence checks. All dynamic thresholds and controls were frozen before outcomes. The failed Qwen3-8B bank is retained as a transparent non-result; the SkillRL final-policy bridge separately records its 18/24 competence calibration, 24 paired A/B/C/D causal units, exact A/D coupling, and the post-negative statistical-resolution audit that prevents upgrading the qualified realization STOP into a population-level no-effect claim.

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